Statistics for Computer Scientists

Lecture 3: Exploring relationships between two variables (bivariate relationships)

19 September 2025

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Where are we

Lecture	Topic	Book
5 Sept	Introduction, statistics fundamentals	Chapter 1
12 Sept	Descriptive statistics	Chapter 2
19 Sept	Bivariate relationships	Chapter 3
26 Sept	Data collection	Chapter 4
10 Oct	Sampling distributions	Chapter 7
17 Oct	Estimation	Chapter 8
31 Oct	Significance testing	Chapter 9
7 Nov	Comparing two groups	Chapter 10
14 Nov	Categorical data analysis	Chapter 11
21 Nov	Linear regression	Chapter 12
28 Nov	Multiple regression	Chapter 13
5 Dec	Tba	
12 Dec	Tba	

Today's lecture

- Recap lecture 2

- Associations between two categorical variables
- Associations between two quantitative variables
- Linear regression
- Confounding variables

Recap

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Recap

- Categorical variables (nominal vs ordinal)
- Quantitative variables (discrete vs continuous)
- Histogram shapes

- Descriptive statistics
  - Center: mean, median, mode
  - Variability: range, standard deviation, IQR
  - Position: quartiles
- Linear transformation and standardizing

Relationships between two variables

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Associations

- Certain values of one variable are more likely to occur with certain values of another variables
  - Height and weight: associated because high weight is more likely at high height
  - Smoking status and lung cancer: associated because higher proportion of lung cancer for smokers
  - Gender and height: associated because men are generally taller than women

- Response variable: outcome variable, dependent variable
- Explanatory variable: “explains” the variation in response variable

Categorical associations

- Survey data on happiness and income
  - Variable types?
- Is there an association between happiness and income?
  - "are certain levels of happiness more likely at certain values of income?"
  - Response and explanatory variable?

		Happiness			
		Very	Pretty	Not too	*Total*
Income	Above average	272	294	49	615
	Average	454	835	131	1420
	Below average	185	527	208	920
	Total	911	1656	388	2955

Contingency table

- Used to show counts of each pair of categories
- If no association, marginal probability equal to conditional probability!
- $P(very\ happy) = \frac{911}{2955} = 0.3$
- $P(very\ happy\ |above\ avg) = \frac{272}{615} = 0.44$

		Happiness			
		Very	Pretty	Not too	*Total*
Income	Above average	272	294	49	615
	Average	454	835	131	1420
	Below average	185	527	208	920
	Total	911	1656	388	2955

Conditional proportions

- Two categorical variables with two categories each
- Interested in proportion of infected people depending on vaccine status
  - $\hat{\pi}_{vac} = \frac{116}{919} = 0.13$
  - $\hat{\pi}_{not} = \frac{35}{81} = 0.43$
- Difference of proportions: $0.43 - 0.13 = 0.30$
- Ratio of proportions: $\frac{0.43}{0.13} = 3.3$
- Significance tests in lecture 7/8

$$\hat{\pi}_{overall} = \frac{151}{1000} = 0.15$$

	Infected	Not infected	Total
Vaccinated	116	803	919
Not vaccinated	35	46	81
Total	151	849	1000

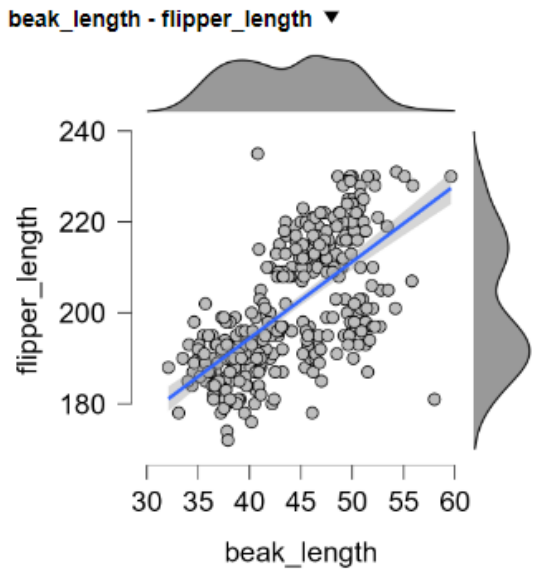
More categories?

- Many comparisons of proportions…
- Statistical test available: chi-squared test of independence
  - Lecture 8!

		Happiness			
		Very	Pretty	Not too	*Total*
Income	Above average	272	294	49	615
	Average	454	835	131	1420
	Below average	185	527	208	920
	Total	911	1656	388	2955

Quantitative variables

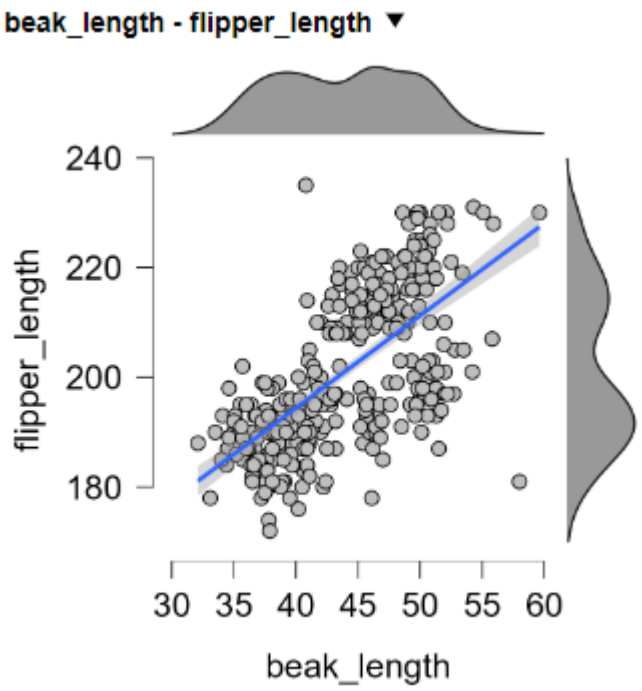
- Penguin dataset
- Is there an association between flipper length and beak length?
- Visual inspection: scatter plot



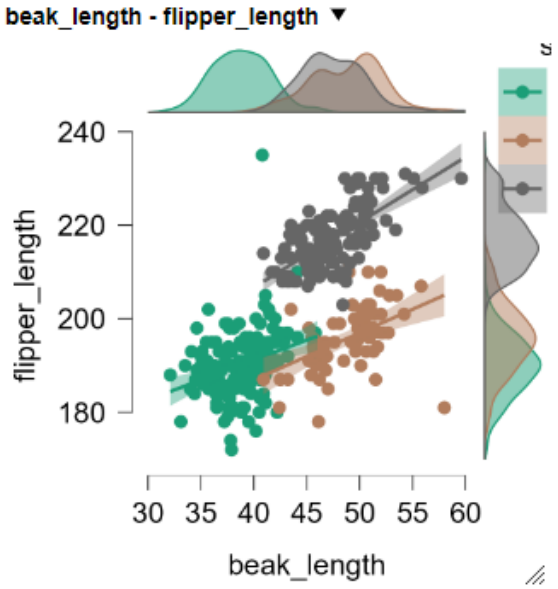
	species	island	beak_length	beak_depth	flipper_length	body_mass	sex
1	Adelie	Torgersen	39.1	18.7	181	3750	male
2	Adelie	Torgersen	39.5	17.4	186	3800	female
3	Adelie	Torgersen	40.3	18	195	3250	female
4	Adelie	Torgersen	39	19.1	194	3300	female

Linear relationships

- Can the trend be approximated by a straight line?
- Is there a positive or negative relationship?
- Are there any unusual observations (far from the trend line)?



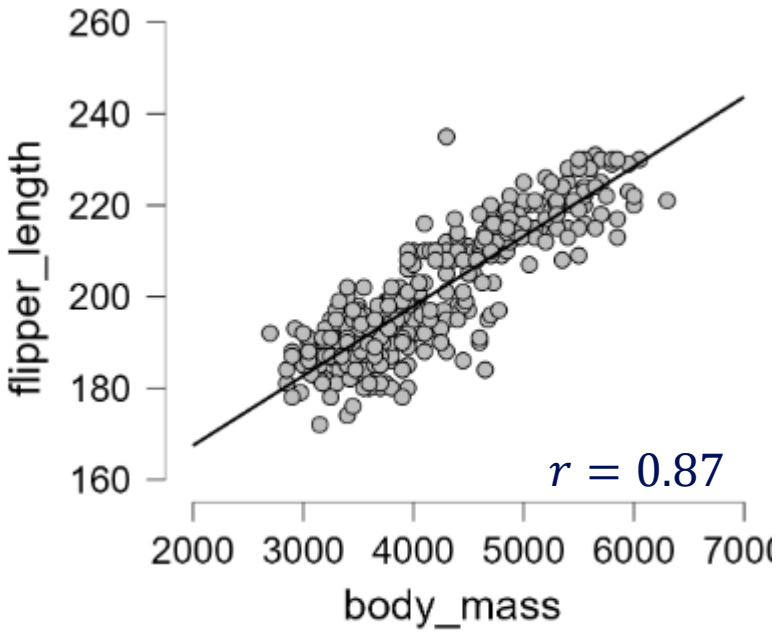
- Watch for clusters!

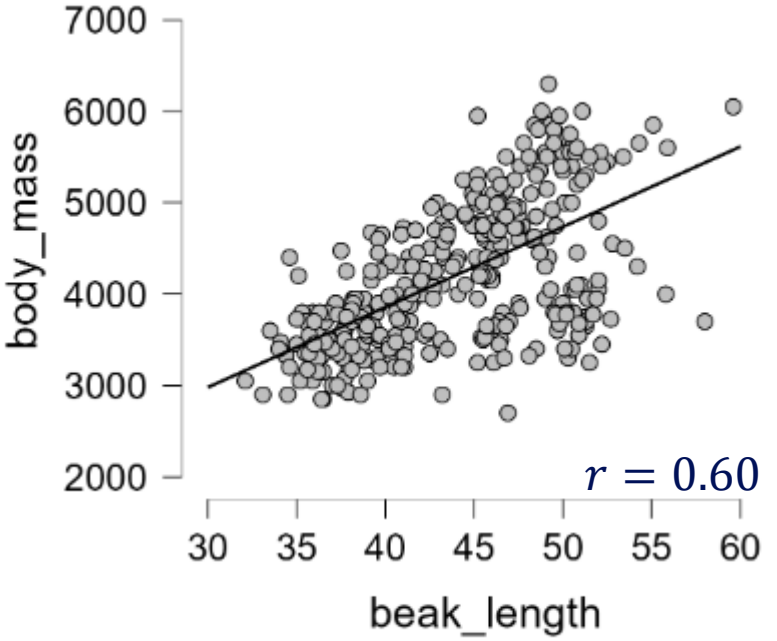


Correlation

- Summarizes the **strength** of the association
- How closely do the data follow the straight-line trend?
  - Not the same as slope!!
- Correlation coefficient r
  - Values between -1 and 1
  - Does not depend on response/explanatory
  - Does not depend on variable units
  - Only appropriate for linear relationships

https://istats.shinyapps.io/guesscorr/





Calculation of correlation

- Book page 156 explains the formula

- Intuition: correlation is a standardized slope
  - Standardizing: correcting for mean and standard deviation
- $r = b\left(\frac{s_x}{s_y}\right)$ where b is the slope of the least squares prediction equation
  s_x and s_y are standard deviations of variables x and y

- Linear regression!

Linear regression

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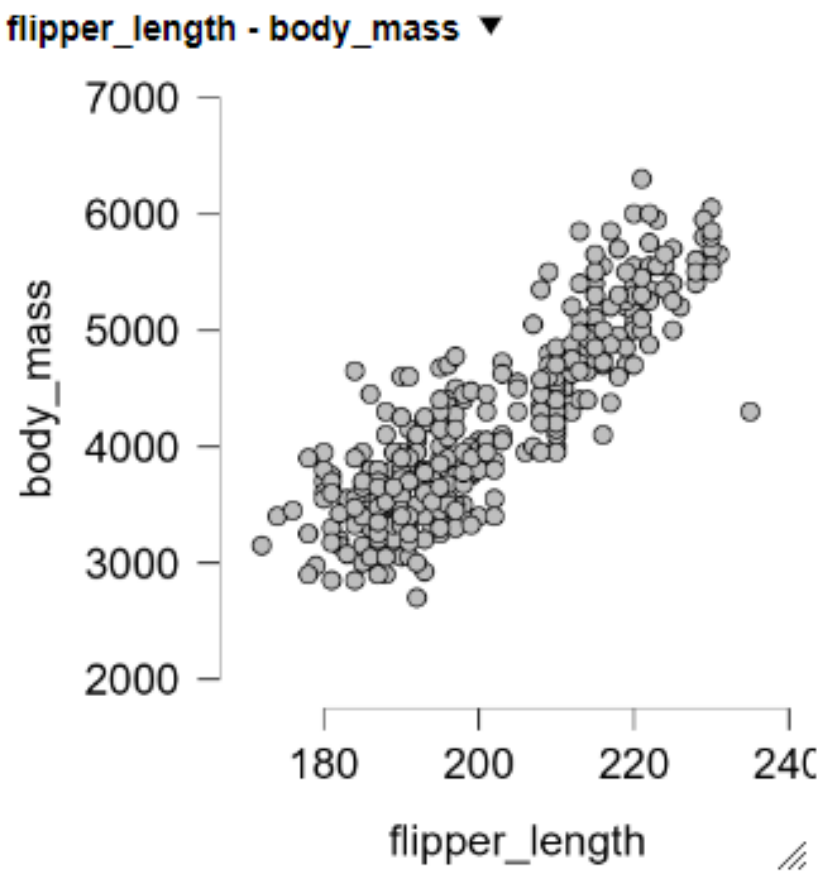
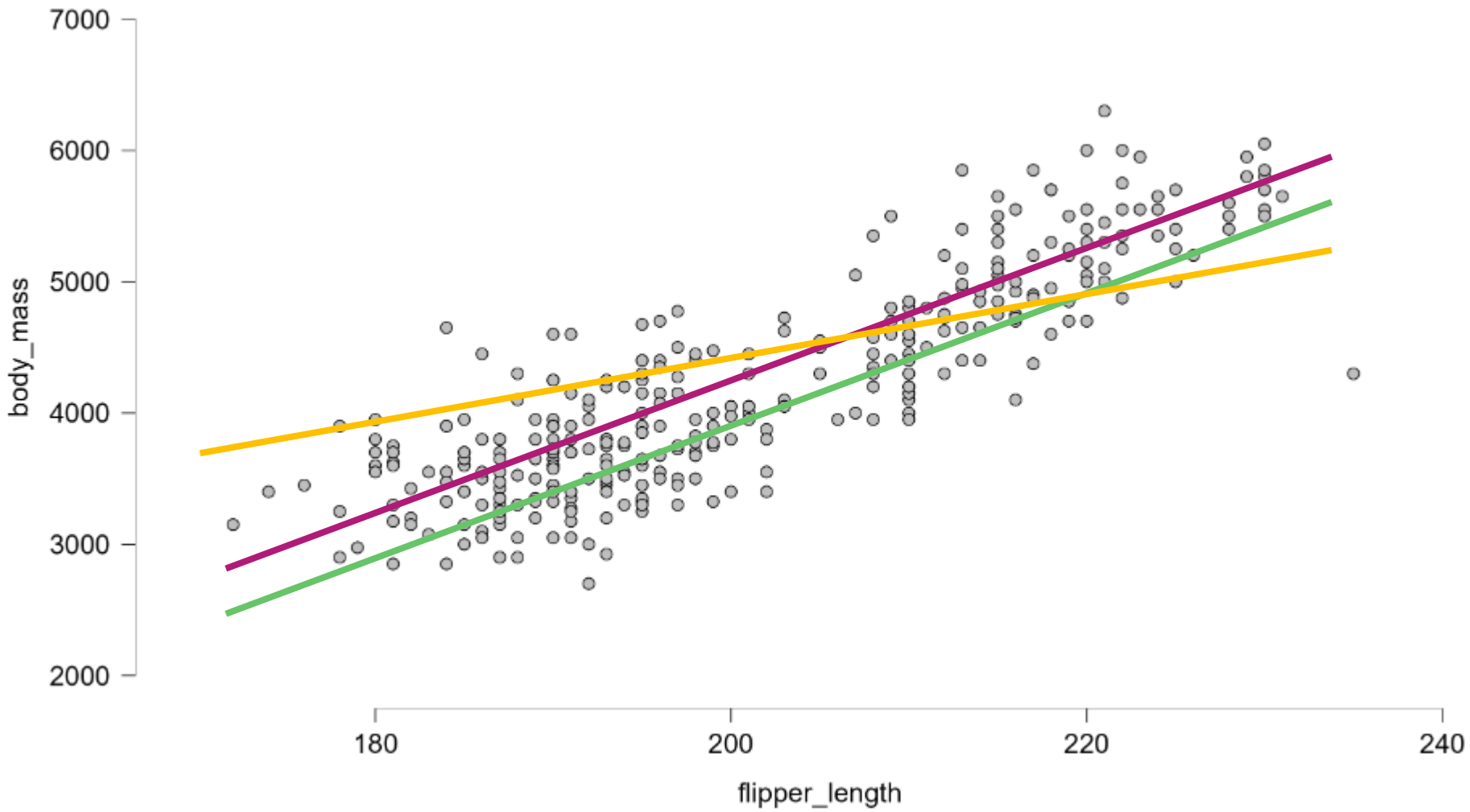
From describing to predicting

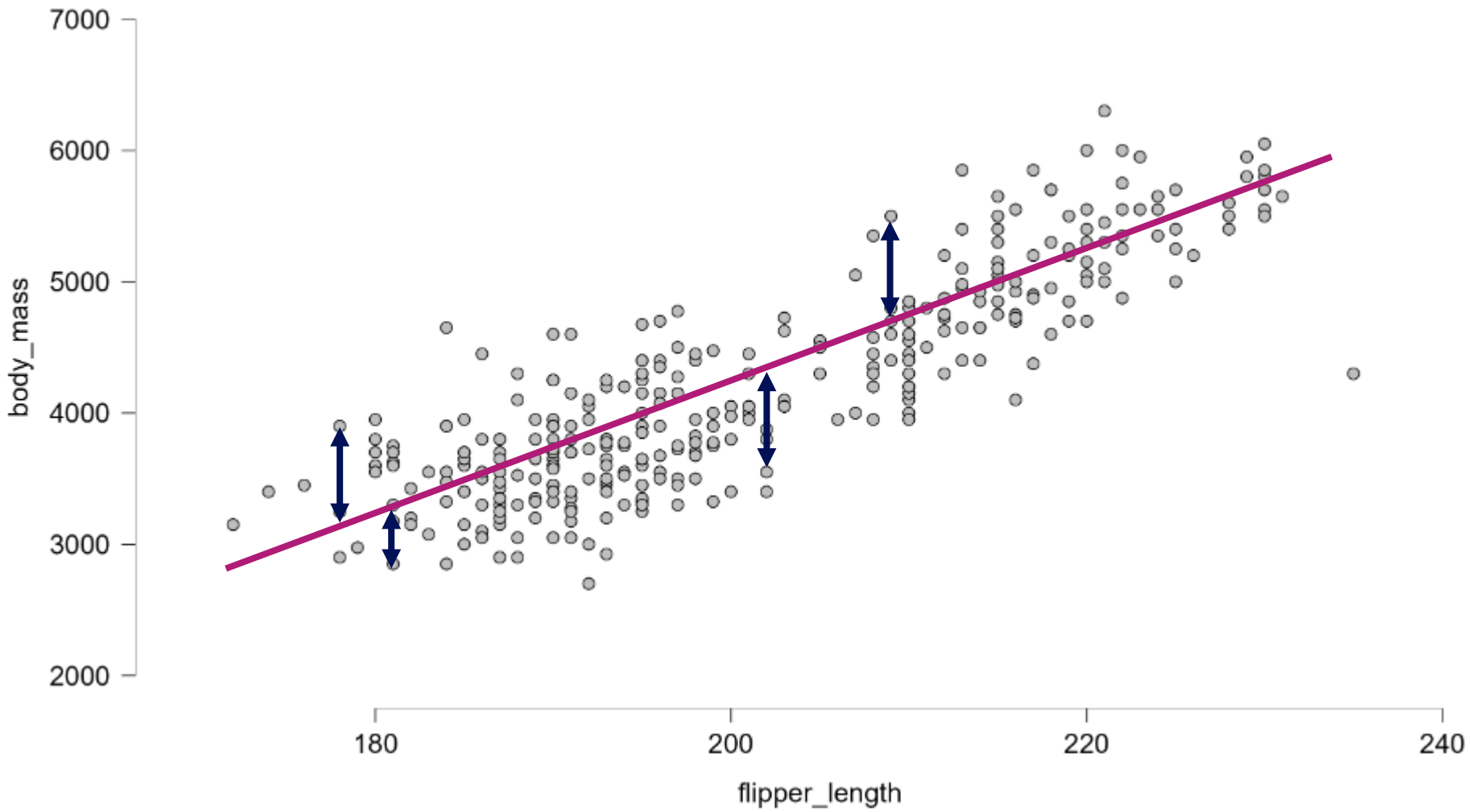
- Scatterplot and correlation describe the association between two quantitative variables
- How to predict Y from X?
- Need to find the best prediction line



Least-squares prediction line



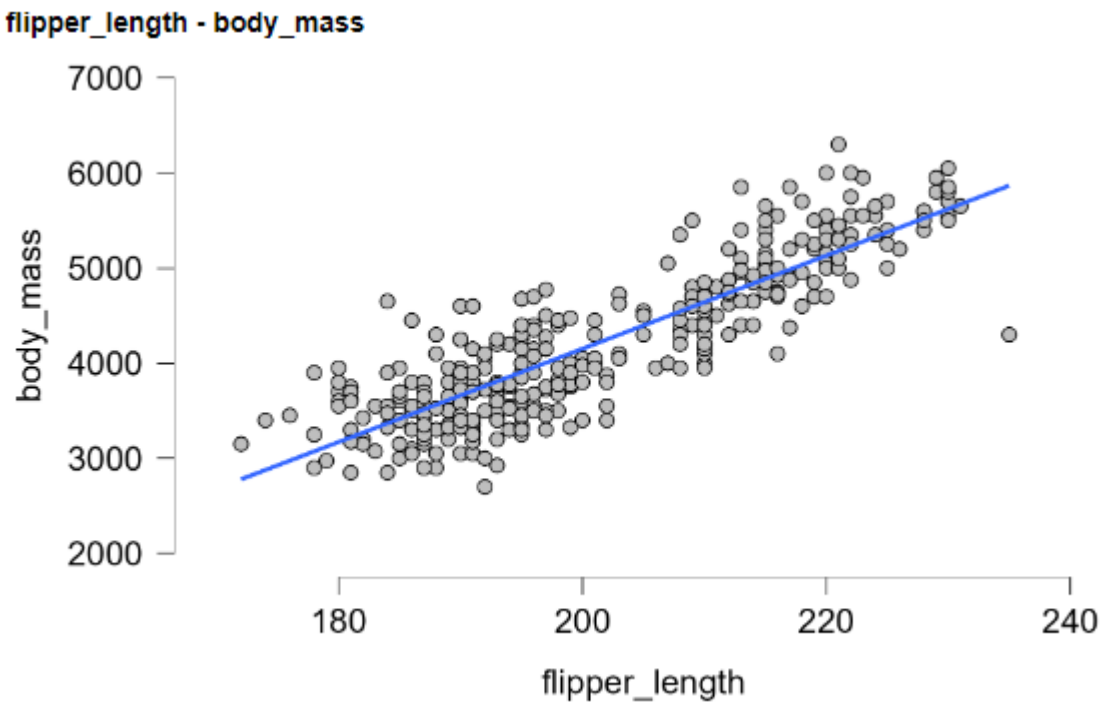
Least-squares prediction line



Prediction equation

- Least-squares prediction line minimizes the sum of the squared differences of each observation to the line (vertically)
  - Difference between predicted and observed!

- Any other line through the data fits worse!



Sample and population

- Population model: $E[Y] = \alpha + \beta x$
- Estimated by prediction equation: $\hat{y} = a + bx$
  - a estimates α, b estimates β, $\hat{y}$ estimates y

- a and b can be estimated from data (usually with software)

$$b = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

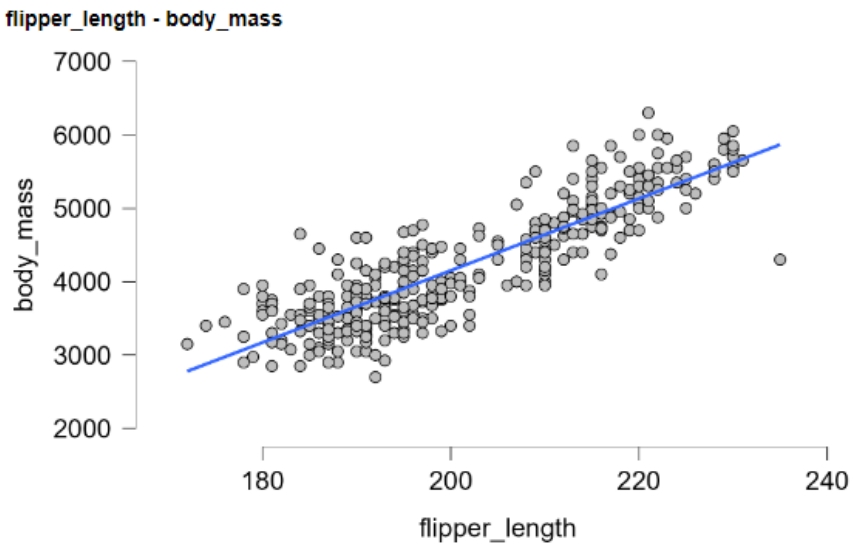
$$a = \bar{y} - b\bar{x}$$

Interpreting $\hat{y} = a + bx$

- Predicting penguin body mass from its flipper length
- $\hat{y} = -5650 + 49x$
- Intercept: the predicted value of y when x is zero (can be hypothetical!)
- Slope: the change in the predicted value of y when x increases by 1

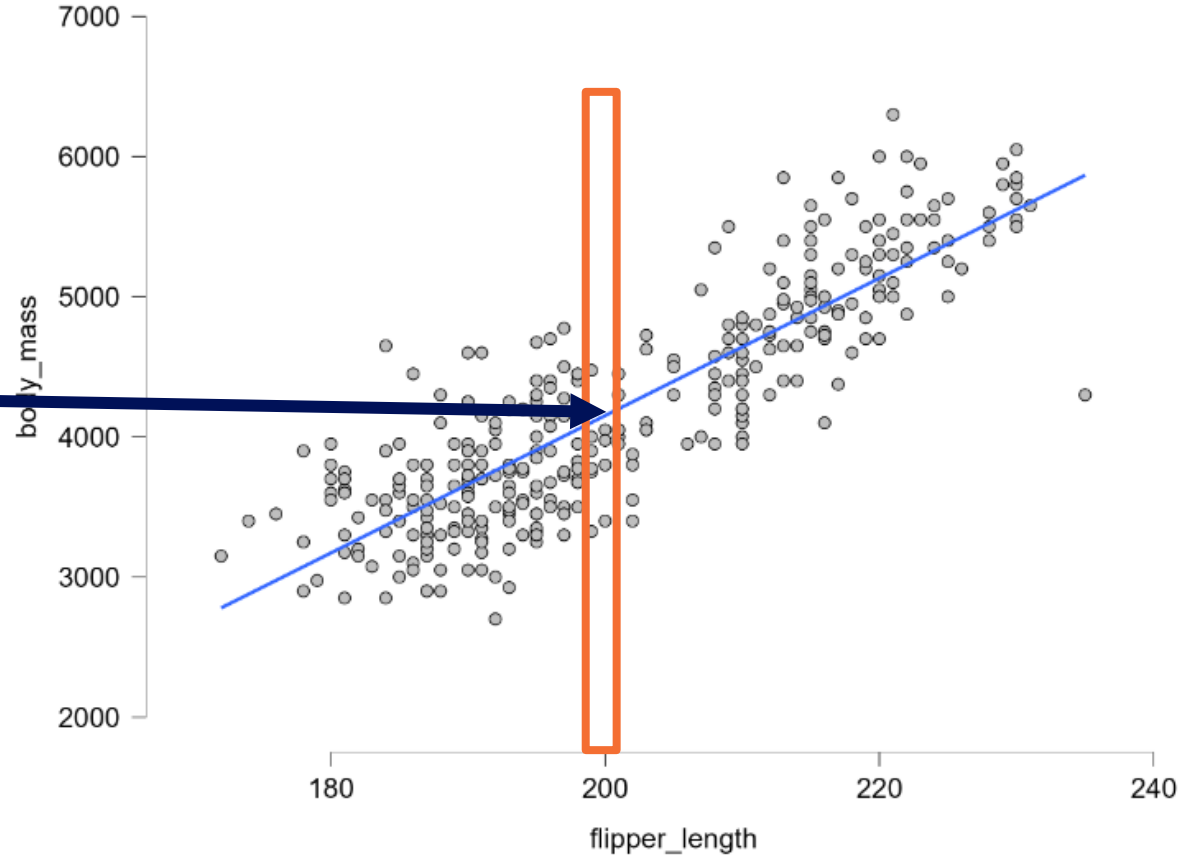
Coefficients

Model		Unstandardized
M_0	(Intercept)	4201.395
M_1	(Intercept)	−5650.112
	flipper_length	49.010



Prediction and variability

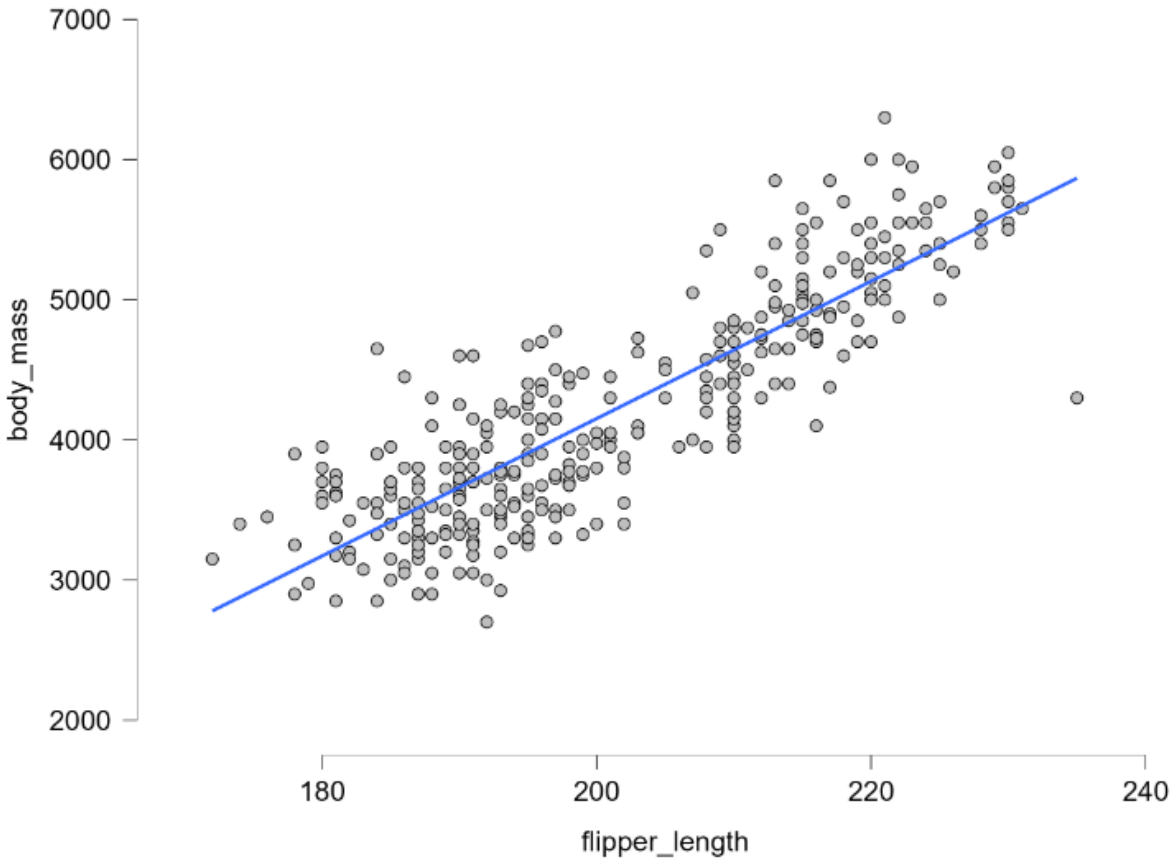
- $\hat{y} = -5650 + 49x$
- Predicted body_mass for flipper_length of 200
  - $\hat{y} = 4150$ cm
- Actual values: 4050, 3975, 3400, …



- In population, the **mean** body mass of penguins with flipper length of 200 is 4150, but there is also variation!

Residuals

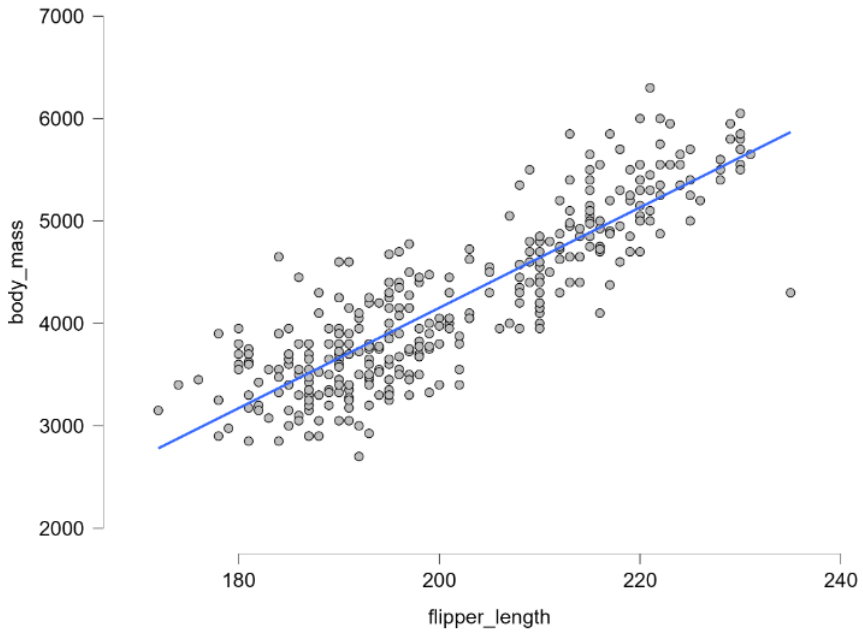
- Residual: difference between predicted and observed value
  - Each observation has a residual
  - $y - \hat{y}$
- Penguin with flipper_length of 200
  - $\hat{y} = 4150$
  - Observed body mass is 4200
  - Residual is $4200 - 4150 = 50$

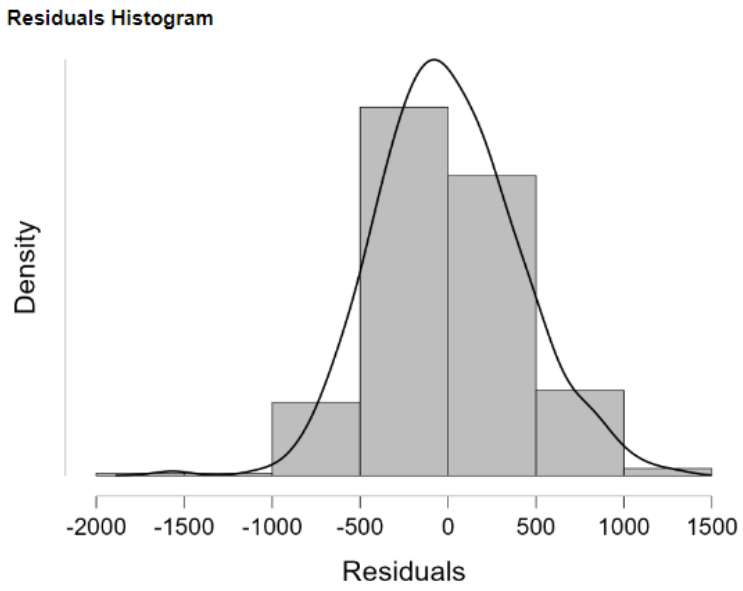


Residuals

- Mean of all residuals is always 0 (because of least-squares line!)
- Inspection of residuals important for regression inference
  - More in later lectures!

- Residual sum of squares
  - $\sum(residual)^2 = \sum(y - \hat{y})^2$
  - Minimized for prediction equation!
  - Lower residual SS is more accurate predictions





r-squared

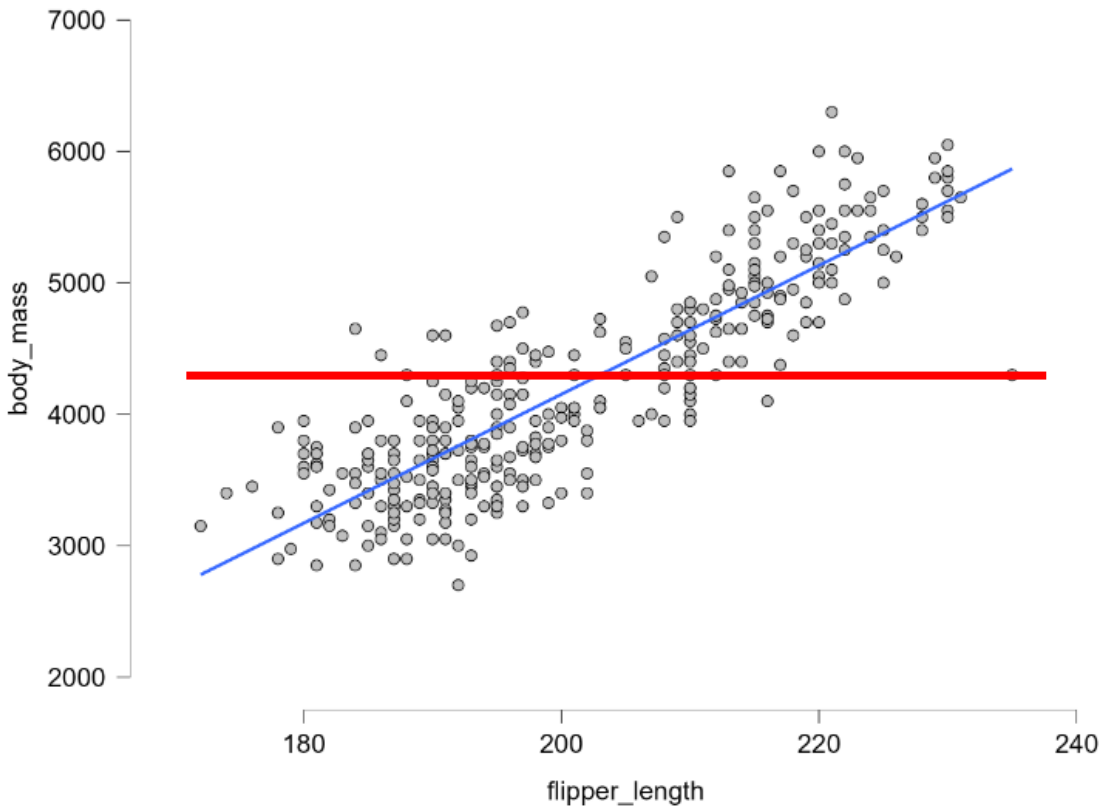
- Assume a quantitative variable Y with $\bar{y} = 50$
  - Best prediction for any observation if we don't know anything else?

- Now assume a quantitative explanatory variable X
- We find $\hat{y} = 10 + 2x$
  - Best prediction for any observation using value of x

- r^2 is the **reduction in prediction error** when using the prediction equation rather than the mean to predict! (also square of correlation)

r-squared

- Residual SS for red line (mean) is larger than residual SS for blue line (pred. equation)
- Observations are closer to the prediction equation than to the mean
- Worst case: prediction equation = mean, then $r^2 = 0$



Model Summary - body_mass

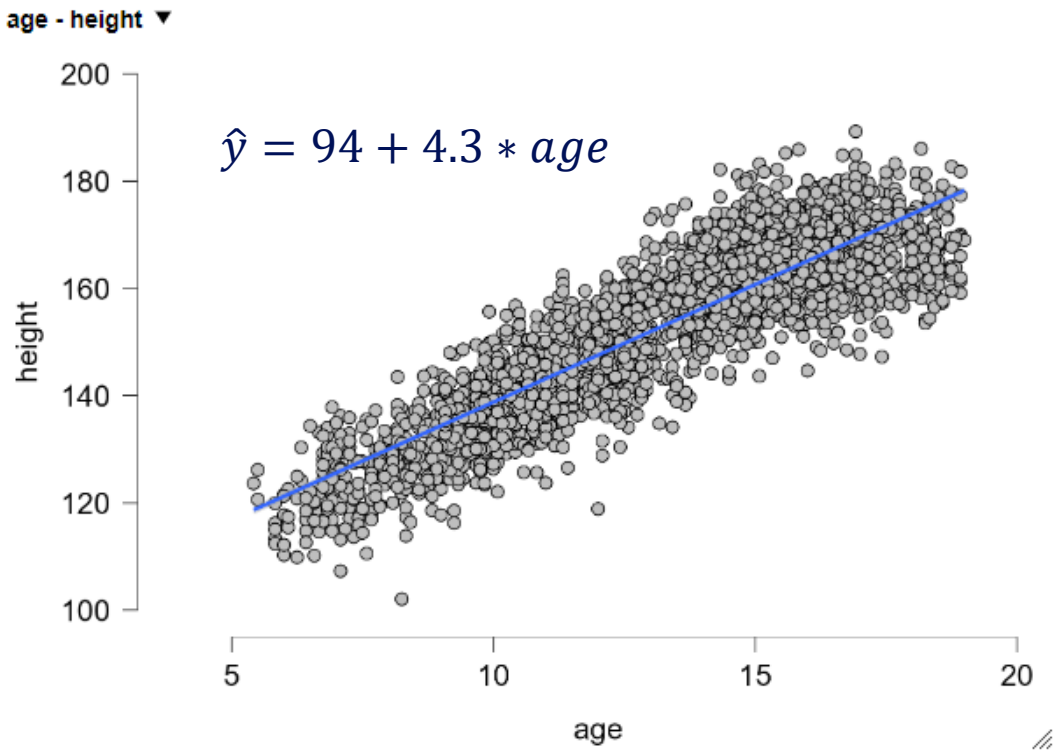
Model	R	R^2	Adjusted R^2	RMSE
M_0	0.000	0.000	0.000	802.195
M_1	0.865	0.748	0.747	403.343

Note. M_1 includes flipper_length

Pay attention to

- Influential outliers: https://istats.shinyapps.io/ExploreLinReg/
  - Value for x is very high or low
  - Observation falls far from the trend

- Extrapolation
  - Prediction equation holds for range of x-values
  - Predicted height for 50-year-old is 309 cm



Pay attention to

- Correlation is not causation!

- Correlation: certain values of y are more likely for certain values of x
- Causation: certain value of x **causes** certain value of y

- You CANNOT conclude causation based on correlation only

https://www.tylervigen.com/spurious-correlations

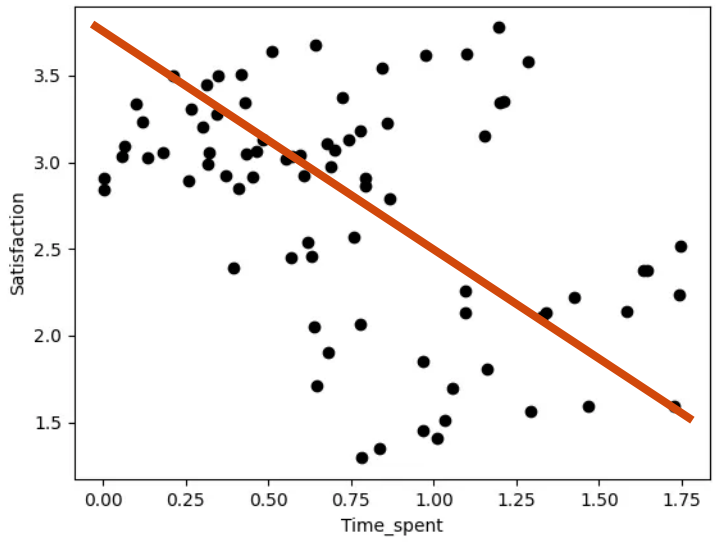
Confounding and lurking

- Lurking variable: not measured in data, possibly confounding
- Confounding variable: available in data, affects relationship between explanatory and response variable

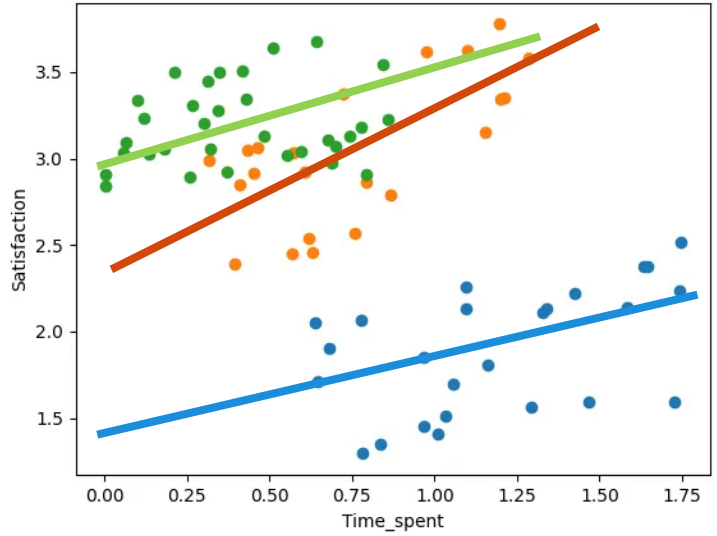
- E.g.: ice cream sales and drowning have a high correlation! Do ice cream sales cause more drowning incidents?

Simpson's paradox

- Describes the phenomenon when the bivariate association is **positive**, yet when controlling for a third variable, the association is **negative** at all values (or vice versa)



- *Example: is spending more time on a certain app associated with people's satisfaction with the app?*
- *Negative association in whole sample*
- *When controlling for country, positive relationship in each group!*



Criteria for causality

1. **Association**

While association does not imply causality, a causal link cannot exist without an association!

2. **Appropriate time order**

Association between X and Y could mean that X causes Y, or Y causes X (or neither)

Criteria for causality

3. No alternative explanation exists

With observational data, the effect of X on Y may be partly due to **lurking variables**: variables that were not observed in the study, but influence either X, Y or both.

With experimental data, we try to control for this by randomization!

Looking ahead: regression

- Today, linear regression with 1 explanatory variable
- Later:
  - Multiple explanatory variables (including categorical)
  - Different sums of squares: total SS, residual SS, regression SS
  - Conditional standard deviation

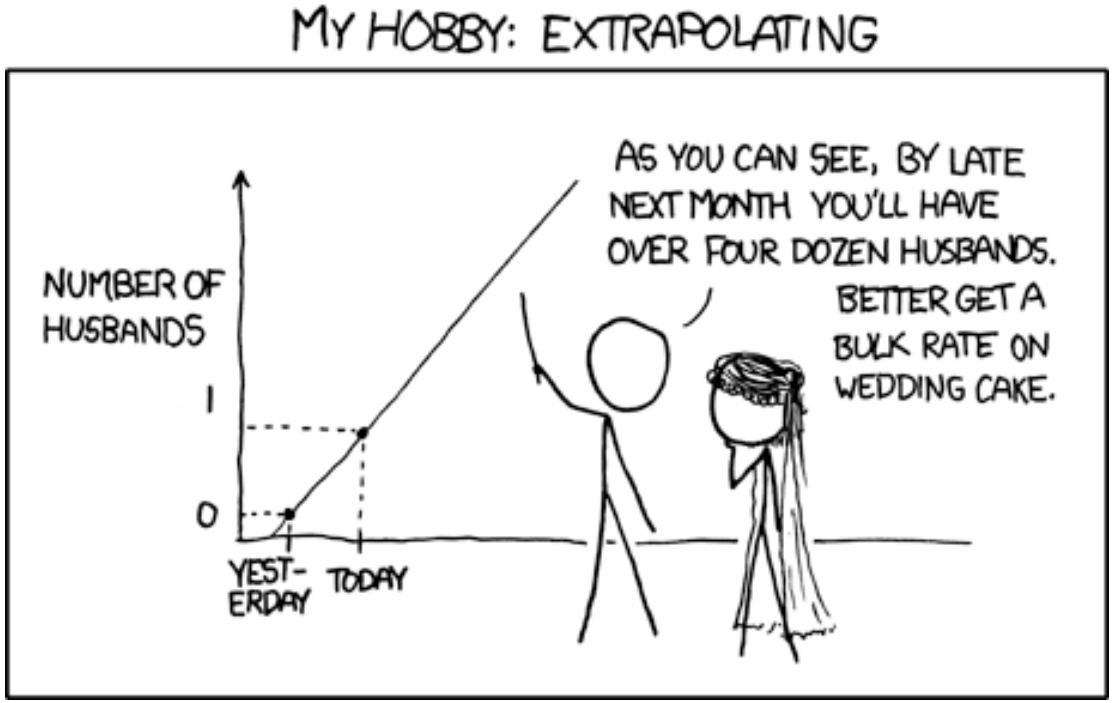
ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	469148.365	1	469148.365	8348.347	< .001
	Residual	140378.998	2498	56.197		
	Total	609527.363	2499			

Wrapping up

- Tutorial: bivariate relationships in JASP
- First JASP exercise online this afternoon, deadline 5 October

- Study chapter 3 in the book
- Brush up on probability if necessary
  - Before October 10



https://xkcd.com/605/